PHENIKAA UNIVERSITY
SCHOOL OF INFORMATION SYSTEMS

COURSE CSE703095
SOFTWARE REQUIREMENTS

SAMPLE REPORT – LAB 08

Safety, RE Tooling & Integration

Topic: Illustrative example of a completed report (reference only, do not copy)

Case study project: MedBook – Online Medical Appointment Booking System
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1 Hazard → NFR Mitigation Cross-check

HZ ID	Hazard	Mitigating NFR	Assessment
HZ-001	Drug interaction warning fails to display	NFR-007 (safety response time), FR-008	**Insufficient**: needs an added NFR on warning accuracy (false-negative rate)
HZ-002	Booking conflict (double booking)	NFR-007	Sufficient, verified by TC-002
HZ-003	Medical record data breach	NFR-003, NFR-004	Sufficient at the design level, needs periodic penetration testing added (not yet in test_cases.csv)

2 Automated Safety Test Results

```
test_TC001_booking_succeeds ... ok
test_TC002_booking_fails_on_conflicting_slot ... ok
test_TC003_cancel_more_than_2h_before_succeeds ... ok
test_TC004_cancel_within_2h_rejected ... ok
test_TC005_drug_interaction_warning ... ok
test_TC005b_no_false_positive_when_safe ... ok
----------------------------------------------------------------------

Ran 6 tests in 0.001s
OK
```

Listing 1: unittest -v test_safety_scenarios.py

3 Self-proposed New Scenario

Situation: a patient tries to book 2 simultaneous appointments with 2 different doctors (no doctor conflict, so NFR-007 as currently defined is not violated). The team discussed: should this be blocked? **Conclusion:** should not be hard-blocked, since there are legitimate cases (same-day

multi-specialty visits), but a **soft warning** should be shown asking the patient to confirm. Added TC-013 to test this soft-warning behavior.

```python
def test_TC013_soft_warning_on_same_time_different_doctor(self):
    """TC-013: do not block, but soft-warn on same-time bookings with
        different doctors."""
    self.sys.book("John Doe", "Dr. B", SLOT)
    result = self.sys.book("John Doe", "Dr. C", SLOT,
       allow_soft_conflict=True)
    self.assertTrue(result.soft_warning)
```

4 Extended Tool: risk-matrix

```
                 Likelihood: Low        Medium       High
Severity High         HZ-001,003          -            -
Severity Medium            -          HZ-002,004       -
Severity Minor             -               -         HZ-005
```

Listing 2: re_tool.py risk-matrix output

5 Overall Project Summary

Across 8 labs, the team built a complete requirements document set for MedBook: 8 stakeholders, 12 FRs, 10 NFRs, 10 use cases, 13 test cases (after additions), 6 change requests, 5 hazards – all tightly linked through the Traceability Matrix and verified by 14 automated tests (7 from Lab 7 + 6 from Lab 8 + 1 extension), achieving 100% PASS. The `re_tool.py` utility enables fast lookup of any project information, demonstrating the feasibility of ”managing requirements as structured data” throughout the lifecycle.

6 Limitations & Future Work

- The dataset is still small (12 FRs) compared to real projects (potentially hundreds of requirements) – future work: integrate a commercial RE tool (Jira, IBM DOORS, Jama).
- NFR simulations use randomized data; should be replaced with real load testing (JMeter/k6) once an actual system is deployed.

7 Self-assessment

Criterion	Max	Self-score	Notes
Hazard → NFR analysis	2.0	2.0	Critical commentary on HZ-001, HZ-003
Safety simulation & testing	2.5	2.5	6/6 PASS
Sound new scenario	2.0	2.0	Well-reasoned business analysis
Extended tooling	2.5	2.0	risk-matrix complete, InsuranceVerifier still simple
Overall presentation	1.0	1.0	
Total	**10.0**	**9.5**	